Unit - 4

Q: Explain photochromic materials, their properties, types of compounds used, and applications. Explain the working principle of photochromic materials.

- Photochromic materials are those that show a reversible change in optical properties (color) through the action of light, i.e., electromagnetic radiation.
- They are transparent materials which exhibit increased light absorption when they are exposed to light.
- Photochromic materials can be in the form of optical glasses, plastic materials, or sometimes delivered as a powder.
- The photochromic properties usually result from the addition of a photochromic substance to a transparent material.

- Photochromic materials and systems have several important uses depending on the rates of the optical transformations:
    - Very slow transformations are useful for optical data storage media.
    - Fast transformations are required for optical switches.

- Photochromic compounds such as:
  - Diarylethene (DE)
  - Dithienylethene (DT)
  - Furylfulgide (FF) derivatives

  are thermally stable and resistant to photochemical side reactions, and were mainly used in optical information storage.

- Photochromic molecules such as:
  - Spirooxazines (SO)
  - Spiropyrans (SP)

  have been embedded into sol–gel glasses to produce optical switches.

<u>working principle of photochromic materials</u>

- The working of photochromic materials is based on a reversible photochemical reaction that occurs when the material is exposed to light.
- The photochromic material absorbs light energy and undergoes a reversible chemical reaction that changes its molecular structure and color.
- When the light source is removed, the material returns to its original state.
- The light-sensitive molecules inside photochromic lenses shift their structure when they are bathed in UV rays, allowing them to absorb more light and give the lenses a darkened appearance.
- Once the UV light goes away, the process reverses and the lenses become clear again.

- These photochromic molecules are sensitive to both the presence and amount of UV light.
- They change their structure more quickly when exposed to a lot of UV light, and more slowly if there's only a little.
- That is why photochromic lenses darken a bit less in the shade than out in the sun.

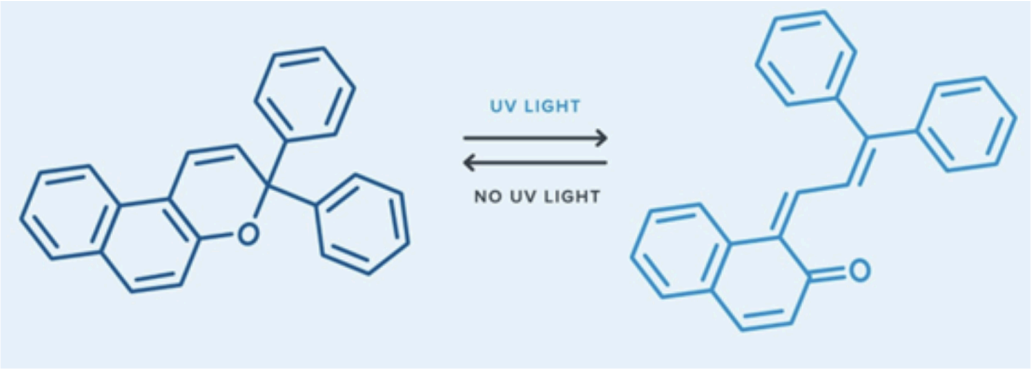
Q: Explain the structure and mechanism of photochromic lenses.

Photochromic lenses are transparent indoors, but darken when exposed to ultraviolet light from the sun.
Essentially, they provide the benefits of glasses and sunglasses in a single frame.
Plastic photochromic lenses have carbon-based compounds within them that change their structure in the presence of UV light.
These organic compounds are also known as photochromic dyes.
Glass photochromic lenses have minute amounts of silver halide crystals (most commonly silver chloride) distributed inside them.
When ultraviolet light hits these chemical compounds, the silver gains an electron and becomes elemental silver—the silver metal with which you're probably familiar.
The lenses appear darker as more and more silver molecules "appear" and absorb visible light.

<u>Structural change of Spiro molecules during UV irradiation</u>

- In the above structure, a napthopyran molecule exposed to UV light undergoes a reversible structural change.

- The ultraviolet light powers a restructuring of the molecule on the left, which is cleaved between the carbon and the oxygen atom.



This principle is used much the same way to make photographic film: exposure to light causes the image to darken and develop.

However, unlike photos, glass photochromic lenses are able to revert back to their original clear appearance.

In the absence of ultraviolet light, a second compound (typically copper chloride) that's also embedded within the glass takes the transferred electron back from the silver metal.

The process reverses, and the lenses become transparent once more.

Advantages of Photochromic Lenses

- The two main purposes of photochromic lenses are to protect your eyes from UV light and to eliminate the need for a separate pair of prescription sunglasses.
- The lenses provide complete protection from UVA and UVB rays at all times.

Disadvantages of Photochromic Lenses

- Photochromic lenses get darkened when coming in contact with sunlight, blocking your windshields. Therefore, they are risky to wear while driving.
- Some photochromic lenses are not polarized, leading to harsh glares of sunlight.

Applications of Photochromic Materials

- Supramolecular chemistry: Photochromic units can be used to create molecules that can switch their shape, function or interactions with light.
- Data storage: Photochromic compounds can be used to store information in optical media, such as CDs or DVDs, by changing their reflectivity or fluorescence with light.
- Solar energy storage: Photochromic materials can be used to store solar energy by converting light into heat or electricity, and then releasing it when needed.
- Smart textiles: Photochromic dyes can be used to make fabrics that change color with light, such as for fashion, art or medical purposes.

Q: What are thermochromic materials? Explain their types with examples and mention both organic and inorganic thermochromic materials.

Thermochromic materials are materials that change color due to a change in temperature. They can be used for various applications, such as temperature indicators, mood rings, baby bottles, kettles, battery testers, and coffee cups.

Types of Thermochromic Materials

Liquid crystals are organic compounds that can reflect different colors at different temperatures, depending on their molecular structure and spacing.

- Examples of liquid crystals include:
  - Cholesteryl esters
  - Cholesteryl ethers
  - Cholesteryl benzoates
  - Cholesteryl oleyl carbonates

These are often used in thermometers, thermal cards, mood rings, and other devices that require precise temperature measurements.

Leuco dyes are organic molecules that can change from colored to colorless or vice versa when they interact with other chemicals at certain temperatures.

- Examples of leuco dyes include:
  - Crystal violet lactone
  - Malachite green lactone
  - Phenolphthalein
  - Fluoran
- These are often used in papers, polymers, inks, and other materials that require a wide range of colors.

Inorganic Thermochromic Materials

- In addition to organic materials, some inorganic materials also exhibit thermochromic properties.
- These include metal oxides such as:
  - Vanadium dioxide (VO_2)
  - Titanium dioxide (TiO_2)
  - Zinc oxide (ZnO)
  - Niobium dioxide (NbO_2)
  - Iron silicide ($FeSi_2$)
  - Titanium sesquioxide (Ti_2O_3)
- These materials change their optical properties due to a change in their crystal structure or electronic configuration at certain temperatures.

Applications:

- mainly used as temperature indicators in many industrial sectors, to determine the temperature variation and reaction heat in chemical reactions.
- used in thermometers for room, refrigerator, aquarium, and medical use, which display different colors at different temperatures.
- serve as indicators of the level of propane in tanks, which change color depending on the temperature and pressure of the gas.
- used in heat transfer mapping devices, which visualize the temperature distribution and convection patterns on a surface.
- used in artistic and educational materials, such as mood rings, thermal cards, stickers, and toys, which change color with body heat or ambient temperature.

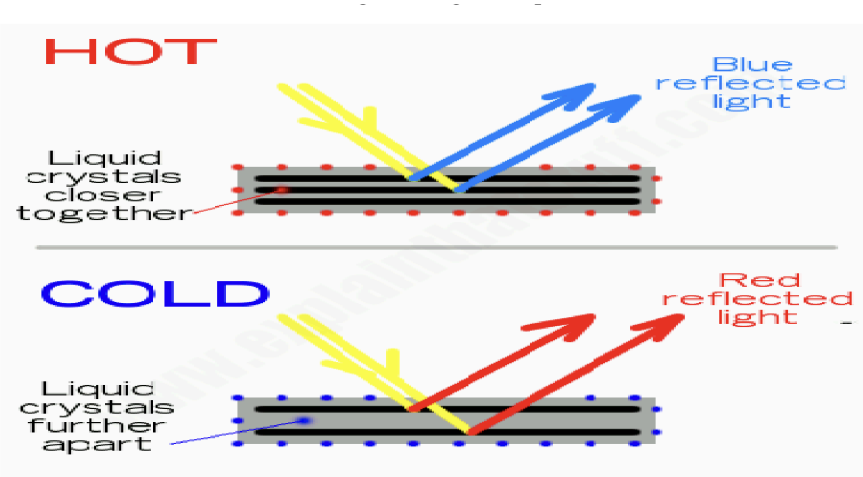
Q: Explain the working and mechanism of liquid crystal thermochromic materials.

Liquid crystal thermochromic materials are materials that change their reflected color as a function of temperature when illuminated by white light.
These materials have chiral (twisted) molecular structures and are optically-active mixtures of organic chemicals.
They can reflect different wavelengths of light depending on the temperature and the orientation of the molecules.

Working Principle

- The principle of working in liquid crystal thermochromic materials is based on the phenomenon of selective reflection of certain wavelengths by the crystalline structure of the material.
- This occurs as the material changes from the:
  • Low-temperature crystalline phase,
  • Through anisotropic chiral or twisted nematic phase,
  • To the high-temperature isotropic liquid phase.
- Only the nematic mesophase has thermochromic properties, which restricts the effective temperature range of the material.

- Liquid crystals change their molecular orientation and spacing when they undergo phase transitions from solid to liquid or from one liquid phase to another.
- This change alters their selective reflection of certain wavelengths of light and produces different colors.



- Top (Hot):
  Incoming light rays hit the layers of liquid crystals (black lines) and reflect back out again, with outgoing rays interfering to produce light of a particular color—in this case, blue.
  The color of the reflected light depends on how closely the crystal layers are together.
- Bottom (Cold):
  Cooling the liquid crystals makes them move further apart, changing the way the outgoing light waves interfere, and making the reflected light redder than before.

Q: What are electrochromic materials? Explain their types, working principle with examples, and list their applications.

Electrochromic materials are materials that can change their color or opacity when an electric field is applied to them. They are different from photochromic materials, which

change color with light. Electrochromic materials can be used to make smart windows, mirrors, sunglasses, and other devices that can control the amount of light and heat passing through them.

Examples of Electrochromic Materials

- Metal oxides: These are inorganic compounds that can change their color by accepting or releasing ions and electrons.
  • eg: tungsten oxide (WO_3) can change from transparent to blue when a voltage is applied.

- Conducting polymers: These are organic compounds that can change their color by changing their oxidation state.
  • eg: polyaniline can change from green to blue when a voltage is applied.

- Organic dyes: These are organic compounds that can change their color by changing their molecular structure.
  • eg: viologen can change from colorless to blue or red when a voltage is applied.

- Among the metal oxides, tungsten oxide (WO_3) is the most extensively studied and well-known electrochromic material.

- Other metal oxides include molybdenum, titanium, and niobium oxides, although these are less effective optically.

Applications of Electrochromic Materials

- Auto-dimming rear view mirrors can reduce glare from other vehicles.
- Smart sunglasses or goggles that can adapt to different lighting conditions.
- Electrochromic displays that can show information with low power consumption.
- Military camouflage gear that can change color to match the environment.
- Sensors, optical shutters or optical modulators, which control the transmission or reflection of light.
- PEDOT:PSS is a polymer mixture of two ionomers based smart electrochromic material.
  - ➔ shows fast, real-time and efficient reversible color change due to the redox process under the influence of the electric field.
  - ➔ The color changes can directly carry readable visual information by the naked human eyes, showing promising applications in smart display, health monitoring, and energy storage.

Q: Explain the principle and device structure of electrochromic materials. Describe the different modes of device operation.

Principle of Electrochromic Materials

- The principle of electrochromic materials is based on the reversible change in their optical properties caused by redox reactions.

- When an electrochromic material is placed on the surface of an electrode, an electric field can induce the redox reactions and change the color or opacity of the material.
- The color change can be between a transparent state and a colored state, or between two colored states.
- The color change can also be controlled by the magnitude of the applied voltage.
- Some electrochromic materials can show several colors, and they are called polyelectrochromic.
- Electrochromic materials can be organic or inorganic, and they can be used to make smart devices that can adjust the amount of light and heat passing through them.

Device Structure of Electrochromic Device (ECD)

- The basic structure of an electrochromic device (ECD) consists of two EC layers separated by an electrolytic layer.
- The ECD works on an external voltage, for which conducting electrodes are used on either side of both EC layers.
- There are three principally different kinds of layered materials in the ECD:
    - The EC layer and ion-storage layer conduct ions and electrons and belong to the class of mixed conductors.
    - The electrolyte is a pure ion conductor and separates the two EC layers.
    - The transparent conductors are pure electron conductors.
- Optical absorption occurs when electrons move into the EC layers from the transparent conductors, along with charge balancing ions entering from the electrolyte.

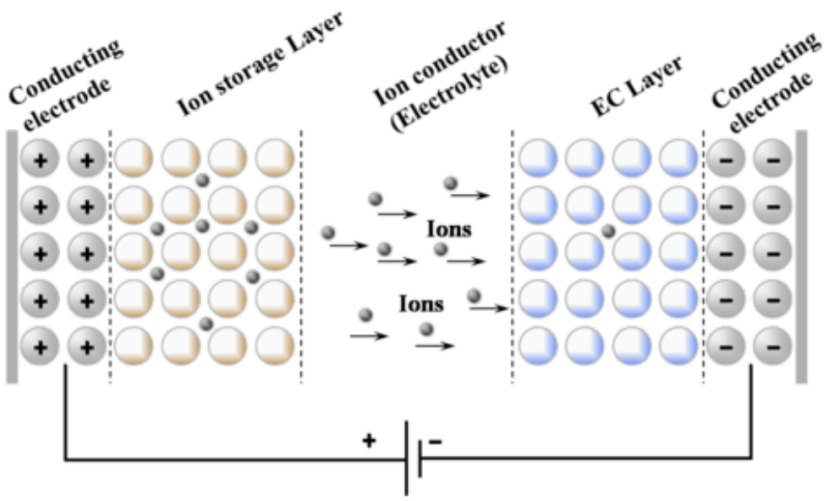


Fig. 4.3- Diagram showing the layers of laminated electrochromic device.

2 Types of ECD Based on Device Operation

 Transmission mode: The conducting electrodes are transparent and control the light intensity passing through them. This mode is used in smart-window applications.

 Reflectance mode: One of the transparent conducting electrodes (TCE) is replaced with a reflective surface like aluminium, gold, or silver, which controls the reflective light intensity. This mode is useful in rear-view mirrors of cars and electrochromic (EC) display devices.

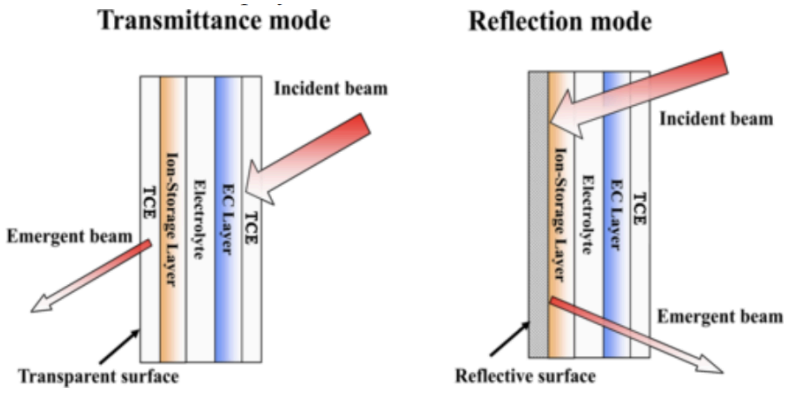


Fig. 4.3.1- Device Operation

Question:
Explain electrostrictive materials with their principle, working mechanism, and applications.

Answer:

- Electrostrictive materials exhibit a quadratic relationship between mechanical stress and the square of the electric polarization.
- Electrostriction can occur in any material, but engineered ceramics called relaxor ferroelectrics have high electrostrictive constants.
- Typical electrostrictive materials:
    - Lead manganese niobate lead titanate (PMN-PT)
    - Lead lanthanum zirconate titanate (PLZT)

<u>Principle:</u>

- These materials change their shape under the application of an electric field.
- This change is caused by the displacement of ions in the crystal lattice, resulting in a compressive force.
- The induced strain is proportional to the square of the applied electric field and does not depend on its polarity.

<u>Working:</u>

- When an electric field is applied, positive and negative ions in the material move in opposite directions.
- This movement creates a compressive force, changing the material's shape.
- The deformation depends on the square of the electric polarization, which refers to the alignment of electric dipoles.
- Electrostrictive materials do not change their shape when the electric field is reversed.

<u>Applications:</u>

- Used in actuator applications. Different types of actuators developed using electrostrictive materials: Hydraulic, Pneumatic, Electrical. Thermal, Mechanical.
- Electrostrictive ceramics are suitable for low-dimensional applications such as:
    - Micro-pumps
    - Positioning systems
    - Micro-electromechanical systems (MEMS)

Question:
Explain magnetostrictive materials with their working mechanism and applications.

Answer:

Magnetostrictive materials are smart materials that convert energy between magnetic and elastic states.
They exhibit magnetostriction — a property that causes them to change shape or dimensions during magnetization.
Examples of magnetostrictive materials:

- Nickel
- Iron
- Cobalt
- Some ferrites
- Rare earths

<u>Working:</u>

- When a magnetic field is applied:
    - Molecular dipoles and magnetic field boundaries (Bloch walls) inside the material rotate to align with the field.
    - This rotation changes the shape and dimensions of the material, causing strain or deformation.

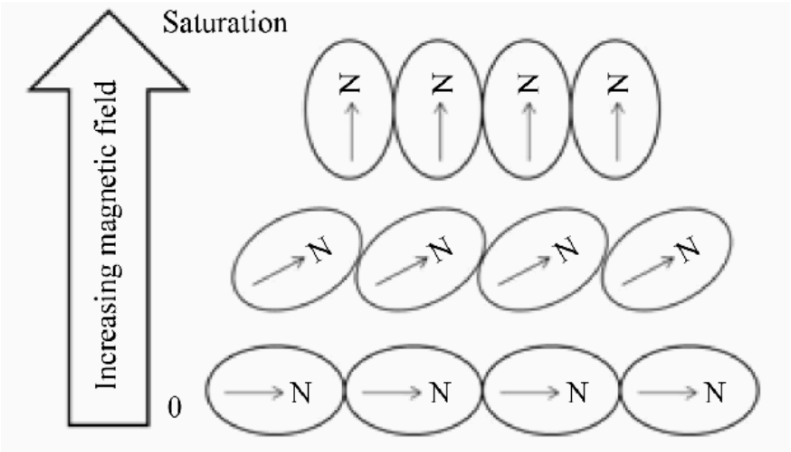
- The amount of strain depends on:
    - Intensity of the magnetic field
    - Magnetic anisotropy (material's tendency to magnetize more easily in certain directions)

- Strain measurement methods:
    - Strain gauges
    - Optical sensors
    - Piezoelectric devices

- Inverse effect:
    - When stress or force is applied, the material’s magnetization changes.
    - This property is used in sensors to measure magnetic fields or detect force.

- If the magnetic field is applied at an angle to an easy axis of magnetization:
    - The structure rearranges to align the easy axis with the field.

- Different crystal directions have different lengths, inducing strain.



Applications:

- Electronic article surveillance – to prevent shoplifting
- Magnetostrictive delay lines – used in early computer memory
- Magnetostrictive loudspeakers and headphones
- Magnetostrictive sensors and actuators – for measuring position, force, and pressure
- Ultrasonic machining – generating high-frequency vibrations for cutting hard materials
- Audio-frequency oscillators – used at sonic and ultrasonic frequencies
- Underwater projectors and sound detectors

Question:
Explain RFID (Radio Frequency Identification) with materials, working mechanism, types, and applications.

Answer:

RFID (Radio Frequency Identification) is a form of wireless communication that uses electromagnetic or electrostatic coupling in the radio frequency portion of the electromagnetic spectrum to uniquely identify an object, animal, or person.

Materials:

- The microchip in an RFID transponder is composed of silicon.
- The antenna can be made of etched copper, aluminum, or conductive ink.
- The chip and antenna are typically placed on a substrate made of Polyethylene Terephthalate (PET).

Working:

- An RFID system consists of three components: A scanning antenna, transceiver, transponder (located in the RFID tag)

- When the scanning antenna and transceiver are combined, they are referred to as an RFID reader or interrogator.

- RFID readers are of two types:
    - Fixed readers
    - Mobile readers

- The RFID reader is a network-connected device that can be portable or permanently attached.

- It uses radio waves to transmit signals that activate the tag.

- Once activated, the tag sends a wave back to the antenna, where it is translated into data.

- The transponder is located in the RFID tag itself.

- RFID tags are made up of:
    - An integrated circuit (IC)
    - An antenna
    - A substrate

- The part of an RFID tag that encodes identifying information is called the RFID inlay.

- The read range for RFID tags varies depending on:
    - Type of tag
    - Type of reader
    - RFID frequency
    - Interference in the surrounding environment or from other RFID tags and readers
    - Tags that have a stronger power source also have a longer read range

<u>Types of RFID Tags:</u>

- Active RFID: Has its own power source, often a battery.

- Passive RFID: Receives its power from the reading antenna, whose electromagnetic wave induces a current in the RFID tag's antenna.

- Semi-passive RFID: A battery runs the circuitry while communication is powered by the RFID reader.

<u>Types of RFID Systems:</u>

- Low-frequency RFID systems:
    - Range: 30 KHz to 500 KHz
    - Typical frequency: 125 KHz
    - Short transmission range: a few inches to less than six feet

- High-frequency RFID systems:
    - Range: 3 MHz to 30 MHz
    - Typical frequency: 13.56 MHz
    - Standard read range: a few inches to several feet
- UHF RFID systems:
    - Range: 300 MHz to 960 MHz
    - Typical frequency: 433 MHz
    - Read range: 25-plus feet
- Microwave RFID systems:
    - Frequency: 2.45 GHz
    - Read range: 30-plus feet

Applications:

- Pet and livestock tracking
- Inventory management, asset tracking, and equipment tracking
- Cargo and supply chain logistics
- Vehicle tracking, access control in security situations
- Shipping, healthcare, manufacturing, and retail sales
- Tap-and-go credit card payments

Question:
What are MEMS (Microelectromechanical Systems)? Explain their structure, size, materials, and manufacturing process.

Answer:

MEMS stands for Microelectromechanical Systems.
It refers to the technology of microscopic devices that incorporate both electronic and moving parts.

MEMS components range in size:

- Between **1 and 100 micrometers** (0.001 to 0.1 mm)
- Entire MEMS devices typically range from **20 micrometres to 1 millimetre** (0.02 to 1.0 mm)

MEMS are composed of the following parts:

- Microsensors
- Microprocessors
- Microactuators
- Data processing units
- Parts that interact with external components

Materials and Manufacturing:

- MEMS devices and integrated circuit (IC) chips are made using similar processes.
- Both begin on a **foundational substrate wafer**, usually made of **silicon** or **glass**.
- Manufacturing steps include:

- **Deposition** – Adding material layers
- **Photolithography** – Patterning the surface through masks
- **Etching** – Subtracting unwanted sections

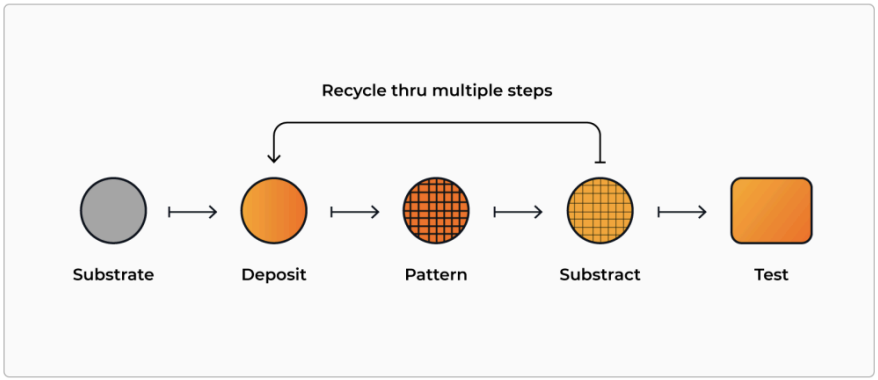


Fig. 4.7.1- Deposition pattern of MEMS

<u>MEMS vs Integrated Circuit (IC) Chips – Key Differences:</u>

1. Material Diversity:
   - MEMS use a wider range of deposition materials.
   - Includes piezoelectrics like lithium tantalate, lithium niobate, and PZT.
   - Also uses noble-metal electrode layers like gold and silver.

2. Complex 3D Fabrication:
   - MEMS fabrication involves additional process steps.
   - Includes deep reactive ion etching (produces near-vertical sidewalls), wafer-level packaging, and very thin layer deposition (<1 micron).

3. Microstructure Shaping:
   - MEMS microstructures are shaped both within the deposited layers and within the substrate.

4. Testing Approach:
   - IC chips are mainly tested digitally — verifying current flow and output.
   - MEMS devices are tested in the physical world, as they are meant to sense or interact with physical parameters.

<u>Applications of MEMS:</u>

- Compact Size:
  - Can be installed in locations where full-size components cannot fit.
  - Essential for smartphones and electronic wearables.
- High Speed:
  - Short electrical distances lead to reduced response time.
- High Performance and Precision:
  - Achieves accuracy levels not possible with conventional components.
- Low Power Consumption:
  - Requires significantly less power than traditional components.

    - Reduces battery load in portable devices.
- High Reliability:
    - Made using silicon materials with low fatigue.
    - Can withstand repeated flexing and sustain extremely long operational lifespans.

Question:
What are NEMS (Nanoelectromechanical Systems)? Explain their features, principle, and applications.

Answer:

NEMS stands for Nanoelectromechanical Systems. These are devices that integrate electrical and mechanical functionality at the nanoscale level.
NEMS represent the next step in miniaturization after MEMS (Microelectromechanical Systems).
They combine transistor-like nanoelectronics with mechanical actuators, pumps, or motors.
NEMS can function as physical, biological, and chemical sensors.

<u>Examples of NEMS Devices:</u>

- Nanoresonators
- Nanoaccelerometers
- Integrated piezoresistive detection devices
- Carbon nanotube nanomotor (produced with hybrid top-down bottom-up fabrication)

<u>Features of NEMS:</u>

- Extremely low mass
- High resonance frequencies
- Operate at nanoscale dimensions
- May exhibit behaviors explained by quantum mechanics, such as:
    - High surface-to-volume ratio
    - Zero point motion

<u>Principle:</u>

- NEMS fabrication is based on:
    - Photo/electron beam lithography
    - Molecular self-assembly
- Comprehensive coverage of MEMS, NEMS, and micromanufacturing is available in the Inspec Database.

<u>Materials:</u>

- Carbon nanotubes are widely used in NEMS:
    - Methods have been developed to connect suspended carbon nanotubes to other nanostructures.
    - Carbon nanotubes help form complex nanoelectric systems.

    - Carbon-based products are controllable and can act as both interconnects and transistors.
    - They are fundamental materials for electrical components in NEMS.
- PDMS (Polydimethylsiloxane) is frequently used:
    - Example: PDMS-coated NEMS diaphragm embedded with silicon nanowires (SiNWs) for detecting chloroform vapor at room temperature.

Fabrication:

- A photoresist is first coated onto the flat surface of the substrate.
- The substrate with photoresist is exposed to a set of lights through a transparent mask containing the desired pattern.
- The patterns on the mask are photographically reduced from macro or mesosizes to the required microscale dimensions.

Applications of NEMS:

- Transportation
- Communication
- Automated manufacturing
- Environmental monitoring
- Healthcare
- Defense systems
- Consumer products
- Sensing
- Displays
- Portable power generation
- Energy harvesting
- Drug delivery
- Imaging

Question:
What is an E-Nose? Describe its definition, sensor materials, and examples used in its construction.

Answer:

An e-nose is a device that can detect and recognize odors and flavors using a sensor array. It uses materials known as chemoresistive materials, which change their electrical resistance when interacting with different molecules.

Materials Used in E-Nose Sensors:

1. Metal Oxides (e.g., tin oxide, zinc oxide, titanium dioxide)
    - Widely used due to:
        - High sensitivity
        - Low cost
        - Easy fabrication

2. Organics (e.g., polymers and carbon-based materials)
    - Advantages:
        - Low operating temperature
        - High selectivity
        - Easy functionalization

3. Two-Dimensional Materials (e.g., graphene, molybdenum disulfide, tungsten diselenide)
    - Features:
        - High surface-to-volume ratio
        - Tunable band gap
        - Fast response and recovery

4. Metal-Organic Frameworks (MOFs)
    - Porous crystalline materials made of metal ions and organic ligands
    - Advantages:
        - High porosity
        - Large surface area
        - Adjustable functionality

5. Other Materials: Includes nanowires, nanotubes, quantum dots, etc., used in e-nose sensors.

<u>Mechanism & Working Principle:</u>

- An electronic nose (e-nose) is a device that detects smell more effectively.
- It consists of:
    - An array of gas sensors that are overlapping selectively
    - A pattern reorganization component
- The e-nose is designed to mimic human olfaction, which functions based on pattern recognition.
- Commonly used sensors in e-nose:
    - Metal oxide semiconductors (MOS)
    - Conducting polymers (CP)
    - Quartz crystal microbalance (QCM)
- Working principle:
    - Based on detection of volatile organic compounds (VOCs) in the air.
    - VOCs are adsorbed onto the surface of the sensor.
    - This causes a change in the sensor's electrical properties.
    - The change is then measured and analyzed using a pattern recognition algorithm.

<u>Applications:</u>

- Used extensively in manufacturing industries for:
    - Quality control and grading
    - Product uniformity and consistency
    - Processing controls

    - Gas leak detection
    - Environmental effluents monitoring

- Other applications:
    - Medical diagnostics and health monitoring
    - Detection of explosives
    - Space applications (e.g., NASA)
    - Research and development
    - Quality control laboratories
    - Process and production departments
    - Detection of drug counterfeiting
    - Food quality-related properties determination

Question:
What is electronic skin (e-skin)? Explain its definition, materials, mechanism, and how it differs from human skin.

Answer:

Electronic skin (e-skin) is a thin, flexible material designed to mimic the function and mechanical properties of human skin.

It contains embedded sensors that can measure: Pressure, Temperature, Humidity, Airflow

E-skin is rehealable due to reversible bond exchange—bonds within the network can break and reform under specific conditions like solvation and heating.
The use of tiny electronic wires enables it to generate impulses, similar to the human nervous system.

<u>Materials:</u>

- E-skin is made from a network of covalently bound polymers that are thermoset (cured at a specific temperature).

- The material is chemically and thermally stable, and also recyclable and reusable.

- Common materials used in tactile e-skin:
    - Carbon nanotubes (CNTs)
    - Organic polymers
    - Graphene
    - Silicon

- Materials used to simulate skin:
    - Liquid suspensions
    - Gelatinous substances
    - Elastomers
    - Epoxy resins
    - Metals
    - Textiles

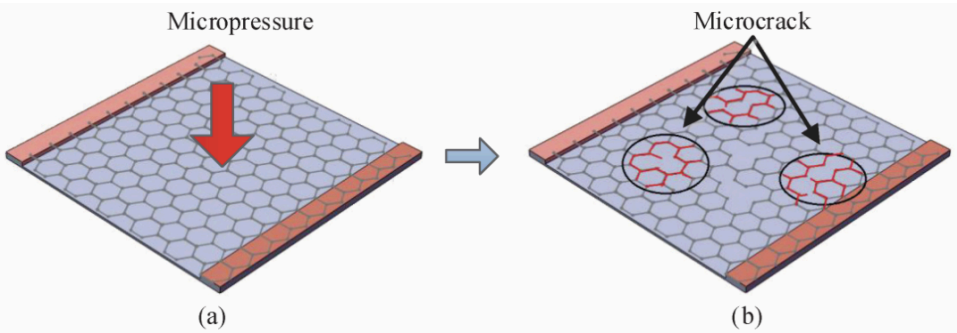
- Nano- and micro-fillers can be added to adjust physical properties of the skin models.

Difference Between Human Skin and E-Skin:

- Human skin is composed of:
    - Epidermis: outer layer; protects from UV radiation and pathogens
    - Dermis: middle layer; contains blood vessels, nerves, and hair follicles
    - Hypodermis: innermost layer; contains fat cells
- Electronic skin (e-skin):
    - Is flexible, stretchable, and self-healing
    - Can mimic functionalities of human or animal skin
    - Has embedded sensors for pressure, temperature, humidity, and airflow
    - Uses tiny electronic wires to generate impulses, resembling the human nervous system

Principle:

- The working of the flexible tactile sensor in e-skin is based on the piezoresistive effect of the graphene film.
- When micro-pressure is applied:
    - The upper bump collects stress and transfers it to the graphene film.
    - This stress fractures or cracks the carbon–carbon (C–C) bonds in the graphene film.
    - As a result, the resistivity of the graphene film changes.
- The change in resistance is used to measure the applied micro-pressure.
- Due to the high sensitivity and flexibility of the graphene film, the sensor achieves high precision and responsiveness.



Common Requirements of E-Skin:

- Heterogeneous integration is needed among components such as:
    - Sensors
    - Active devices
    - Interconnects
    - Communication interfaces
    - Energy management circuits
    - Data storage and processing units

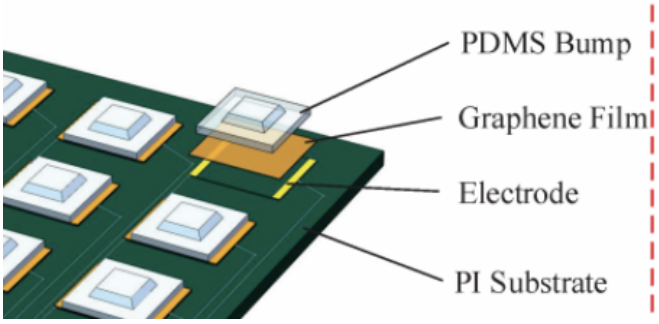
    - Energy harvesting components

- Key requirements for all components:
    - Biocompatibility
    - Conformability to skin topography
    - Flexibility
    - Stretchability to various degrees
    - Sustainability under body motion strain
    - Minimal discomfort during wear
    - Reasonable adhesion to skin
    - Adequate shelf life and service life
    - Lightweight design

<u>Materials:</u>

composed of 4 × 4 tactile sensing units, each having three layers:

1. Lower substrate – Polyimide substrate (PI substrate)
2. Piezoresistive layer – Graphene/polyethylene terephthalate (PET) film
3. Top bump layer – Polydimethylsiloxane (PDMS)



<u>Mechanism:</u>

- The piezoresistive curve of e-skin has two segments based on applied pressure:

  Segment 1: 0 kPa to 4 kPa
    - Corresponds to small pressures
    - PDMS bump deforms, pressing the graphene film
    - This generates many small cracks in the graphene film
    - Due to the low Young's modulus of PDMS, large deformation occurs with small stress
    - The resistance of the sensor increases noticeably

  Segment 2: 4 kPa to 500 kPa
    - Corresponds to larger pressures
    - PDMS bump deformation saturates
    - PET substrate now begins to deform
    - This causes additional small cracks in the graphene film
    - Due to the high Young's modulus of PET, more pressure is required

- Sensitivity decreases in this range

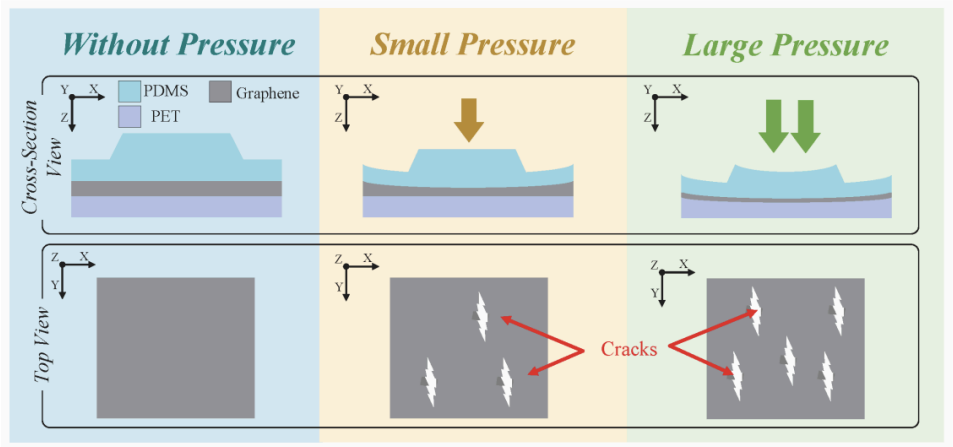


Fig. 4.10.2- Mechanism of the electronic skin

<u>Applications of E-Skin:</u>

- Health monitoring and wireless mechanical sensing.
- Used in prosthetics that mimic the sense of touch.
- Monitors life signs such as:
  - Respiration rate
  - Body temperature
- Used in:
  - Wound care
  - Drug delivery systems
- Applicable in:
  - Wearable devices
  - Skin-attachable electronics
  - Robotics
  - Prosthetics
- Continuous blood glucose monitoring:
  - Important for diabetics to track blood sugar changes
  - Uses sensors with specific enzymes to measure glucose concentration
- Speech recognition system (ASR):
  - An e-skin device attached to the human throat
  - Detects weak pressure changes from muscle movement
  - Transforms signals into speech
  - Helps the deaf and mute communicate

Question:
Define e-waste. Explain its types, associated environmental risks, and the importance of proper recycling.

Answer:

Electronic waste (e-waste) refers to electronic products that are: Unwanted, broken, or nearing or exceeding their useful life

Examples of common items producing e-waste:

- Computers
- Televisions
- VCRs
- Stereos
- Copiers
- Fax machines

E-waste includes:

- Outdated, end-of-life, or abandoned electronic appliances
- All associated parts, consumables, and spares

The issue of disposing used electronics dates back to at least the 1970s.

<u>Types of E-Waste:</u>

- Type 1 – Major appliances
  - Refrigerators
  - Washing machines
  - Dryers

- Type 2 – Small appliances
  - Vacuum cleaners
  - Irons
  - Blenders
  - Fryers

- Type 3 – Computer and telecommunication appliances
  - Laptops
  - Personal computers (PCs)
  - Telephones
  - Mobile phones

<u>Environmental Risks:</u>

- E-waste is not harmful if: Stored safely, recycled scientifically, or transferred properly through the formal sector

- E-waste becomes harmful when:
  - Recycled using primitive technologies
  - Toxic compounds from metals in e-waste leak into the environment, especially when buried
  - Informal or uncontrolled processing: Common in developing countries & causes pollution and poses serious health risks to humans and the environment.

Question:
List the major toxic compounds found in e-waste, their sources, and associated health effects.

Answer:

Toxic Compounds in E-Waste:

Compound	Sources	Health Effects
Lead	Monitor gaskets, glass panels, solder components, PCBs	Affects nervous system, kidneys, reproductive and endocrine systems; impairs brain development in children.
Cadmium	SMD chip resistors, infrared detectors, semiconductors, old CRTs	Accumulates in the body, mainly in kidneys; toxic.
Mercury	Thermostats, sensors, relays, switches, lamps, batteries, medical equipment, phones	Damages brain, kidneys, fetus; converts in water to dimethyl mercury—bioaccumulates in food chain.
Chromium VI	Steel plating, decoration, corrosion protection	Destroys DNA; highly toxic to the environment.
PVC & Plastics	Cable insulation, computer housings	Burning emits dioxins; toxic fumes.
Barium	CRT front panels	Short-term exposure causes brain swelling, muscle weakness, and organ damage.
Beryllium	Motherboards, finger clips (Cu-Be alloy)	Causes lung cancer, skin conditions; effects may appear years after exposure.

Toners	Found in black/color printer cartridges	Inhalation causes respiratory irritation; carbon black is a possible human carcinogen (Class 2B).
Phosphor & Additives	Coating inside CRT faceplates; includes cadmium, zinc, vanadium	Highly hazardous; risky for workers dismantling CRTs manually.

Question:
Briefly explain the environmental and human health impacts of e-waste.

Answer:

<u>Impacts on Environment:</u>

- Manufacturing, recycling, and disposal of e-waste pollutes air, water, and soil.
- Burning wires and cables releases brominated and chlorinated dioxins, polluting air.
- Toxic substances with low commercial value are discarded, causing contamination.
- Effluents are sometimes discharged into subsurface aquifers, leading to groundwater pollution.
- Dismantling activities release dust containing heavy metals and flame retardants.
- Dust particles redeposit nearby or travel far, polluting land and water systems.
- Dumping lead, mercury, cadmium, arsenic, and PCBs in landfills causes soil pollution.

<u>Impacts on Human Health:</u>

- Poor e-waste handling leads to chemical exposure and inhalation, especially in developing nations.
- Workers lack protective equipment and are often unaware of the risks.
- Burning PCBs raises dioxin levels, increasing cancer risk.
- Handling acids and metals during metal extraction leads to toxic exposure and blood contamination.
- Burning insulated wires causes neurological damage.
- Cadmium exposure harms kidneys, liver, and bones.
- Long-term lead exposure from screens and PCBs damages the nervous system and kidneys, especially in children.

Question:
Explain the recycling and recovery process of e-waste. Mention key steps and related global guidelines.

Answer:

Recycle Management Process:

1. Collection:
    - E-waste is gathered from enterprises, organizations, and individuals.

2. Storage:
    - Important for hazardous components like CRT glass (contains lead).
    - Due to low CRT demand, such glass is now stored indefinitely.

3. Manual Sorting, Dismantling, and Shredding:
    - Components are disassembled to identify reusable parts.
    - E-waste is shredded into small pieces for better material separation.

4. Mechanical Separation:
    - Consists of multiple sequential processes:

5. a) Magnetic Separation:
    - A large magnet extracts ferrous metals (iron, steel).
    - Eddy current may be used to separate non-ferrous metals.
    - Also separates metal-embedded plastics and circuit boards.

6. b) Water Separation:
    - Separates plastics and glass using water.
    - Aims to further purify and hand-sort remaining contaminants.

7. Recovery:
    - Separated materials are prepared for reuse or resale.
    - Some materials enter other recycling streams (e.g., plastic, steel).
    - Others may be processed and sold on-site.

Extraction of Valuable Metals:

- E-waste is hazardous but contains precious metals (PMs) including:
    - Gold (Au), Silver (Ag), Platinum (Pt), Palladium (Pd),
    - Gallium (Ga), Tantalum (Ta), Tellurium (Te),
    - Germanium (Ge), Selenium (Se)

Global Impact and Regulatory Guidelines:

- Rapid electronic growth and short product life cycles have made e-waste the fastest-growing segment of municipal waste.

- Basel Convention:
    - Regulates transboundary movement and disposal of hazardous waste.

- RoHS Directive (Restriction of Hazardous Substances):
    - Limits use of toxic materials like:
        - Lead (Pb), Mercury (Hg), Cadmium (Cd), Chromium (Cr),

        - Polychlorinated biphenyls
    - Promotes substitution with safer alternatives, reducing environmental harm and load on recycling plants.

- E-waste generation is growing at 5–10% annually, making proper recycling essential.

QUIZ: https://revisely.com/quiz/x2LWOr & https://revisely.com/quiz/BnI3E9